Introduction to Research Study Designs and Evidence in Dentistry

Dr Faiza Siddiqui

Medical Educationist DDC

OBSERVATIONAL STUDIES

Learning Objectives:

- By the end of this session, students will be able to:

1.Define key types of research study designs.

2.Differentiate between observational and experimental studies.

3.Appreciate the importance of evidence-based dentistry.

4.Apply knowledge through a group-based classification activity.

Observational study Design : Case Report/Series

- What it does?

Detailed description of 1 or few cases

Example

Rare oral tumor in teen

Observational study Design : Cross sectional

- What it does?

Snapshot of a population at one point in time

Example

Survey on brushing habits

Observational study Design :
Case control

- What it does?

Compares patients **with** a condition to those **without** (retrospective)

Example

Oral cancer vs non-cancer patients: smoking history

Observational study Design : Cohort

- What it does?

Follows people over time to see who develops an outcome (prospective or retrospective)

Example

Track kids with braces to see if they develop gum issues

Quick review
Observational studies

Design	What It Does	Example
Case Report/Series	Detailed description of 1 or few cases	Rare oral tumor in teen
Cross-Sectional	Snapshot of a population at one point in time	Survey on brushing habits
Case-Control	Compares patients with a condition to those without (retrospective)	Oral cancer vs non-cancer patients: smoking history
Cohort	Follows people over time to see who develops an outcome (prospective or retrospective)	Track kids with braces to see if they develop gum issues

Experimental Studies

Experimental Studies:
Randomized Controlled Trial (RCT)

- What it does?

Participants are randomly assigned to different treatments

Example

Compare 2 types of dental sealants

Experimental Studies: Non-Randomized Trial

- What it does?

Similar to RCT but no random assignment

Example

Assign sealants based on school location

Quick review
Experimental studies

Design	What It Does	Example
Randomized Controlled Trial (RCT)	Participants are randomly assigned to different treatments	Compare 2 types of dental sealants
Non-Randomized Trial	Similar to RCT but no random assignment	Assign sealants based on school location

Syntheses / Reviews

Systematic Review

- What it does?

Collects and critically analyzes multiple studies

Example

All RCTs on fluoride varnish

Meta-Analysis

- What it does?

Combines results statistically

Example

Combines cavity reduction rates from 10 studies